

Jenny Liu

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EDUCATION

Branksome Hall, Toronto, Ontario, Canada

Sep 2022 - Present

- HL Math AA, HL Physics, HL Chemistry, HL Economics, SL Spanish, SL English Language and Literature
- Academic honors: Recognition of Outstanding Attitude and Effort in Mathematics, Recognition of Outstanding Effort and Achievement in Computer Studies, Recognition of Outstanding Effort and Achievement in Visual Arts, Recognition of Outstanding Effort and Achievement in Mandarin

HONORS & AWARDS

Gold Medal, Toronto Science Fair

Mar 2024

Excellence Award, Canadian Intermediate Mathematical Competition (CIMC), Top 5%

Nov 2023

Excellence Award, Canadian Senior Mathematical Competition (CSMC), Top 5%

Nov 2024

Silver Medal, Drawing & Illustration Category, Scholastic Arts and Writing Competition

Dec 2023

Excellence Award, National Young Artist Portrait Competition

Nov 2023

RESEARCH EXPERIENCE

5th Author of research publication: "Analysis of the Driving Stability of a Multi-Model Drive Electric-Tracked Intelligent Detection Device," published in Actuators (MDPI), 2024, Vol. 13(12), SCI Expanded. Apr 2024 - Nov 2024

- Attended a 3-month college-level research internship program at Jilin University and co-authored a 9000+ word research paper under the advisory of Professor Zhenglei Yu, Jilin University.
- Researched and drafted literature review after reading 25+ hours per week; developed virtual prototype simulation model with RecurDyn and Matlab/Simulink; Collected forward, vertical, and roll angular acceleration data for stability analysis for three operating conditions with the physical prototype.
- Link to publication: <https://www.mdpi.com/2076-0825/13/12/527>

Independent Researcher, Tribology: "Ultralow friction of hydrogenated DLC films-based lubricating coating for space application in low earth orbits"

May 2023 - Aug 2023

- Attended the independent research Eureka program and authored a 5000+ word research paper, advised by Professor Sadaf Sobhani, Cornell University.
- Analyzed hydrogen DLC films' chemical composition, optimized hydrogen content for space missions in low Earth orbit under atomic oxygen and UV exposure, and proposed lifespan improvements through dopant incorporation.
- Acquired an A+ for the course, with my paper nominated for publication in Eureka Journal 2023.

Independent Researcher, Emergency Door Release Mechanism in Vehicle Fires

Nov 2023 - Mar 2024

- Gold medal project at Toronto Science Fair.
- Designed a prototype of the door release component with Solidworks; conducted 13 tests on its reliability with a total of 65 trials under safety supervision; analyzed data with Python and used ANOVA test to draw conclusions.
- Shared research experience with 500+ students at STEM assembly.
- Link to project board: https://www.canva.com/design/DAGADokzkRw/f9CEYalk9HYIzCptCDGF8w/edit?utm_content=DAGADokzkRw&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton.

Independent Researcher, Industrial Design, Drinking Fountain Redefined

Nov 2022 - Jan 2023

- Utilized Inventor and Blender to redesign the drinking fountain to enhance accessibility and user experience at school; Added stretching tubes, and sterilization lamps around the water outlet, and redesigned the appearance of the water dispenser.
- Presented at Branksome Hall to 100+ students and parents; earned an A+ for the personal project showcase.

INTERNSHIP EXPERIENCE

Valiant TMS, Windsor, Canada

Jun 2024 - Jul 2024

Intern at Industrial Design Department and Coding Department

- Created 5 rough molds from drawings for an existing project with Inventor; numbered all the project's drawings according to specifications; participated in and documented 3 design department meetings with external suppliers of conveyor belts, shaft collars, and flywheels.
- Coded 100+ lines with TP programming language and spent 20+ hours observing Fanuc robots in action on the floor.

EXTRACURRICULAR ACTIVITIES

Robotics Club

Sep 2024 - Present

Club Head

- Organized and led meetings with 30+ robotics club members for 6 hours weekly across 3 days.
- Participated in 5 Vex competitions in Toronto with 10 team members each time.
- Coordinated the team's participation in 2025 Vex Girl-Powered Online Challenges; Interviewed a female electrical engineer and created a 3-minute video empowering girls to thrive in robotics.

World Affairs Conference Technology Team

Sep 2024 - Present

Technology Head

- Led weekly technology meetings with 10+ members; presented bi-weekly updates to 40+ WAC attendees.
- Developed a website with 1500+ lines of code using JavaScript, HTML, and CSS, integrated with Firebase for student-teacher registration, event details, participant management, and more.
- Invited Dr. Roberta Bondar, Canada's first female astronaut, to serve as the keynote speaker for the World Affairs Conference, attended by 1,200+ participants from 15+ Toronto schools.

STEM Council

Sep 2023 - May 2024

Grade 10 STEM representative, Assembly Host, Organizer

- Organized project-based and game-based learning activities like frog dissection, STEM clubs fair, and “craft your planet” with a total of 500+ participants.
- Hosted a STEAM assembly for 1000+ students and delivered a speech on my science exploration.

International Student Association

Sep 2023 - Present

Club Head

- Advocate for international cultures to 500+ participants from 20+ countries at the International Student Assembly.
- Organize the largest mentorship program at school, gathering 120+ participants annually.
- Host bi-weekly meetings with 25+ students to foster cultural understanding and address various dilemmas.

Perennial (Student-led Art Magazine)

Oct 2022 - Present

Chief Art Editor

- Publish 2 arts and writing editions per year, featuring 10 literature writing and 10 artworks by students.
- Reach out to industries to produce 200+ physical copies
- Post-digital magazine on the school website, gathering overall 50K+ reads and 1K+ likes.

Art Council Representative

Sep 2024 - Present

Grade 11 Representative

- Designed a poster for the Spirit Week Coffee House, featuring 10 performances from the student community and 50+ audience members from junior school during community time.
- Organized the school's Arts Assembly; created advertisements for the drama production "Emma."

Psychology study on teenage mental development in immigrant families

Apr 2024 - Oct 2024

Team Leader

- Interviewed high school students and their parents from 50+ families.
- Designed questionnaires with 15 questions; visualized and analyzed data using Python; generated a physical report with 1000+ words and presented a project report to 200+ students, parents, and teachers.

Personal Interests and Skills:

Computational skills:

- Proficiency in various computing languages: Python (700+ lines), Javascript (500+ lines), HTML (300+ lines), CSS (300+ lines), TP programming language for Fanuc Robot (100+ lines) and Matlab (200+ lines).

Design skills:

- Skilled in using 3ds Max, Inventor, RecurDyn, Blender in visualizing planet models of my interest and prototyping 5 industrial designs during company internships, of 3 eventually employed in final design

Organizational and Communication skills:

- Experience of organizing team progress for robotics club weekly meetings in google doc, real industrial projects' design specifications in excel and inter-school collaboration with Upper Canada College students in github.